

Multigas Calibrator HORIBA APMC-370

Booklet



HORIBA's APMC-370 is designed to calibrate gas analyzers manually, remotely controlled or automatically, installed in air pollution monitoring stations, for quality assurance in the laboratory and also for the production of gas analyzers.

Concept:

Dilution of gases with high accuracy and precision requires high quality mfc's, especially for high dilution ratios and ranges. Consequently HORIBA's APMC-370 is equipped with digital mfc's. These mfc's are factory calibrated with a special Polynomial Calibration

Technical Features:

- Latest generation of Calibrators
- Fulfills the requirements of the latest EN-Norms specially in respect to the linearity check
- Exceeds the US EPA requirements
- Measurement methods according to international standards
- Microprocessor controlled with touch screen LCD display
- Precision of span gas concentration : better than 1%
- Linearity of span gas concentration : better than 1%
- Multipoint Calibration : 8 individually adjustable cycles, each with upto 20 calibration points
- Each calibration cycle runs also automatically
- Operation, manually or via V24, Ethernet or dry contacts
- Both 19" rack mounted version or Portable version
- Options :
 - a. Additional mass flow controllers (8 max.)
 - b. Ozone oven for the calibration of O3 analyzers and/or gas phase titration (GPT)
 - c. Internal zero gas calibration system
 - d. Permeation Oven

Technical Description:

A special feature of **HORIBA's** APMC-370 is the big size touch screen panel, for ease of operation. Characteristic of the operation of **HORIBA's** APMC-370 is the intuitive, simple and user friendly menu. (Flow rate, mg/m³, ppb/ppm, automatic cycles etc.) Via the touch screen it is e.g. possible to enter span gas concentrations or to start automatic routines like multi point calibration cycles.

Routine sequences like calculating and adjusting span gas concentrations as well as complex flow patterns for multi point calibration cycles can be stored and easily activated. All functions are controlled by an internal CPU with a state-of-the-art 32-bit operating system. This internal microprocessor continuously calculates and displays the final gas concentrations in the selected units (ppb/ppm or mg/m³-µg/m³). Adjustment, monitoring and regulation of flow rates are done fully automatically. These cycles are programmed in the unit's memory, thus time spent setting up calibrations on-site is minimized.

All critical operational parameters are monitored; any failure message will be displayed and transmitted via the interfaces.

Integrated data logger: During multi point calibration cycles, the span gas concentrations switch automatically with the integrated data logger stores continuously the data of the connected gas analysers. Afterwards data can be transferred to a standard browser via Ethernet connection for further evaluation.

HORIBA's APMC-370 is a modular system and can be adapted and upgraded any time to accommodate most of customer's requests.

TECHNICAL DATA:

<u>Mass-Flow-Controller (MFC):</u>	
Accuracy of MFC:	<i>+/-0.5% of the actual value plus +/-0.1% of full scale</i>
Reproducibility of MFC:	<i>better than 0.1% of the reading</i>
Repeatability of MFC:	<i>better than 0.2% of the reading</i>
Linearity of MFC:	<i>+/-0.1% full scale</i>
Pressure Sensitivity:	<i>0.1% / bar typical for N2</i>
Inlet Pressure:	<i>1 bar</i>
Total Flow Rate:	<i>approx. 0.1 – 5.2 l/min max.</i>
Accuracy of Span Gas:	<i>better 1%, varying flow rate of one MFC</i>
Temperature Range (ambient air):	<i>5 – 40 degC</i>
Humidity (ambient air):	<i>max. 85%, non-condensing</i>
Interfaces:	<i>2 x RS232, eg. Geystec Protocol (others on request), 1 x 10/100 MBit Ethernet, dry contacts</i>
Display:	<i>TFT 640 x 480 Pixel with touch screen feature</i>
Connectors:	<i>Zero Gas 6mm PVDF, Span Gas 3mm PVDF, Stainless Steel and/or inch connectors (1/4" & 1/8") on request</i>



Principle of Operation:

- Dynamic generation of span gas from gas cylinders, dilution of common gases

Gas Flow Regulation – Dilution Gas:

- Mass-Flow-Controller, Standard Range: 0.1 – 5 l/min (others on request)

Gas Flow Regulation – Span Gas 1:

- Mass-Flow-Controller, Standard Range: 4 – 200 ml/min (others on request)

Options:

Gas Flow Regulation – Span Gas 2 (for higher dilution ratio):

- Mass-Flow-Controller, Standard Range : 0.4 ml – 20 ml/min (others on request)

Depending on the application, mfc's will switch automatically between span gas 1 & span gas 2, eg. For linearity checks over a wide range.

Power:	<i>230V AC +/-10%, 50 Hz (others on request), 50 VA</i>
Dimensions:	<i>430 x 120 x 400 (W x H x D) with brackets for 19" rack mounted version</i> <i>520 x 530 x 200 (W x H x D) for transportable version</i>
Option:	<i>Telescopic Rails for 19" rack mounted version</i>
Weight:	<i>approx. 12 – 14 kg (for 19" rack mounted version)</i> <i>approx. 23 – 25 kg for transportable version</i>

*Due to continuous product improvement, specifications contained herein are subject to change without notice